

Zahid Ullah

Seoul, South Korea

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Education

Changwon National University (CWNNU)

Changwon, Republic of Korea

PhD in Computer Engineering

Mar 2016 - Aug 2020

- **Thesis Title:** An Improved Methodology for Brain MRI Image Enhancement and Classification
- **Supervisor:** Prof. Su-Hyun Lee
- **Research Interest:** Medical Image Analysis using Deep Learning, Artificial intelligence, Computer Vision, Image Processing (i.e., CT, MRI, X-rays)
- **Featured Courses:** Statistical Natural Processing, Network Design, Image Processing, System Analysis and Design, Advanced Algorithm, Advanced Distributed Multimedia Systems

Shaheed Zulfiqar Ali Bhutto Institute of Science and Technology (SZABIST)

Islamabad, Pakistan

Master of Science in Computer Science

Jan 2013 - Jun 2015

- **Featured Courses:** Digital Image Processing, Advanced Computer Architecture, Theory of Computation, Advanced Operating System, Advanced Topics in Artificial Intelligence, Information Security, Software System Architecture

University of Malakand, (UoM)

Khyber Pukhtunkhwa, Pakistan

Bachelor of in Information Technology

Sep 2007 - Dec 2011

- **Featured Courses:** Operating Systems, C++, Discrete Mathematics, Design and Analysis of Algorithm, Network Security, Object-Oriented Programming
- **Final Year Project:** Developed a Website in PHP for Gandhara College, Chakdara

Experience

Dongguk University

Seoul, Republic of Korea

Assistant Professor

September 2024 - Present

- Teaching graduate courses and graduate thesis committee member
- Working on the stitching/registration, and tracking of scalp hair analysis
- Cardiac Segmentation
- KL-Grade Classification
- Brain Tumor Segmentation
- Brain Tumor Classification
- Continual Learning

Korea National University of Transportation (KNUT)

Chungju, Republic of Korea

PostDoctorate Researcher

September 2020 - August 2024

- Teaching graduate courses and graduate thesis committee member.
- Conduct extensive research and development in AI and medical image processing.
- Provide supervision and guidance to master's and Ph.D. students.
- Guided interns working on deep learning-based projects.

Changwon National University (CWNNU)

Changwon, Republic of Korea

RESEARCH Assistant

March 2016 - June 2018

- Development of a hybrid image enhancement based brain MRI images classification technique
- Proposed Enhanced feature extraction technique for brain MRI classification based on Haar wavelet and statistical moments
- Proposed Features Reductions Using Color Moments and Classification of Brain MRI Using K-NN
- Proposed Critical Analysis of Brain Magnetic Resonance Images Tumor Detection and Classification Techniques. (All the projects were supervised by Prof. Su-Hyun Lee)

Teaching Assistant

Sep 2018 - June 2020

- Managing projects - Timeline, workflow and documentation
- Performing Feasibility analysis and literature review.

Research Supervised

MS Thesis

- Completed in co-supervision = 01 student
- In progress = 03 students

KNUT, South Korea
March 2021-Dec 2022

Journal Publications

Primary Authored Publications

- **Ullah Z.** & Kim J. (2025). **Anatomically accurate cardiac segmentation using Dense Associative Networks.** *Engineering Applications of Artificial Intelligence*, **IF: 8.0**.
- **Ullah Z.** & Kim J. (2025). **VL-OrdinalFormer: Vision Language Guided Ordinal Transformers for Interpretable Knee Osteoarthritis Grading.** *Mathematics*, **IF: 2.07**.
- **Ullah Z.**, Hilal M., Lee E., Pamucar D., & Kim J. (2026). **Unified Review and Benchmark of Deep Segmentation Architectures for Cardiac Ultrasound on CAMUS.** *arXiv*.
- **Ullah Z.** & Kim J. (2025). **Hierarchical Deep Feature Fusion and Ensemble Learning for Enhanced Brain Tumor MRI Classification.** *Mathematics*, **IF: 2.07**.
- **Ullah Z.**, Hong M., Mahmood T., & Kim J. (2025). **Systematic Integration of Attention Modules into CNNs for Accurate and Generalizable Medical Image Classification.** *Mathematics*, **IF: 2.07**.
- **Ullah Z.** & Kim J. (2025). **Exploring Kolmogorov–Arnold Network Expansions in Vision Transformers for Mitigation of Catastrophic Forgetting in Continual Learning.** *Mathematics*, **IF: 2.07**.
- **Ullah Z.**, Usman M., Jeon M., & Gwak G. (2022). **Cascade multiscale residual attention CNNs with adaptive ROI for automatic brain tumor segmentation.** *Information Sciences*, **IF: 8.2**.
- **Ullah Z.**, Usman M. & Gwak G. (2023). **MTSS-AAE: Multi-task semi-supervised adversarial autoencoding for COVID-19 detection based on chest X-ray images** *Expert Systems With Applications*, 216, 2023. **IF: 8.5**.
- **Ullah Z.**, Usman M., Latif S., & Gwak G. (2023). **Densely Attention Mechanism-based Network for COVID-19 Detection in Chest X-rays.** *Nature Scientific Reports*, 13, 261. **IF: 4.9**.
- **Ullah Z.**, Usman M., Latif S., Khan A., & Gwak G. (2023). **SSMD-UNet: semi-supervised multi-task decoders network for diabetic retinopathy segmentation** *Scientific Reports*, 10(1), 9087. **IF: 4.9**
- **Ullah Z.**, Farooq M U., Lee S H., & An D. (2020). **A hybrid image enhancement based brain MRI images classification technique.** *Medical Hypotheses*, 143, 109922. **IF: 4.7**
- **Ullah Z.**, Fayaz M., & Lee S H. (2019). **Enhanced feature extraction technique for brain MRI classification based on Haar wavelet and statistical moments,** *International Journal of Advanced and Applied Sciences* 6(7), 2313. **IF: 0.22**

Co-authored Publications

- Ahmad S., **Ullah Z.**, & Gwak J. (2024). **Multi-teacher cross-modal distillation with cooperative deep supervision fusion learning for unimodal segmentation,** *Knowledge-Based Systems*. **IF: 8.8**
- Farooq M U., **Ullah Z.**, & Gwak J. (2023). **DC-AAE: Dual channel adversarial autoencoder with multitask learning for KL-grade classification in knee radiographs.** *Computers in Biology and Medicine.*, **IF: 7.7**
- Farooq M U., **Ullah Z.**, & Gwak J. (2023). **Residual attention based uncertainty-guided mean teacher model for semi-supervised breast masses segmentation in 2D ultrasonography.** *Computerized Medical Imaging and Graphics*. **IF: 7.4**
- Kang J., **Ullah Z.**, & Gwak J. **MRI-Based Brain Tumor Classification Using Ensemble of Deep Features and Machine Learning Classifiers.** *Sensors* 2021. **IF: 3.9**
- Mahmood T., Ullah Z., Latif A., Sultan B., Zubair M., **Ullah Z.**, Ansari A., Zehra T., Ahmad S. & Dilshad N. **Computer-Aided Diagnosis in Spontaneous Abortion: A Histopathology Dataset and Benchmark for Products of Conception.** *Diagnostics* 2025. **IF: 3.0**

Under-review Manuscripts

- **Ullah Z.**, Pamucar D., & Kim J. (2025). Hybrid Ensemble Approaches: Optimal Deep Feature Fusion and Hyperparameter-Tuned Classifier Ensembling for Enhanced Brain Tumor Classification. Under-review at *Engineering Applications of Artificial Intelligences (EAAI)*, **IF: 7.5**.
- **Ullah Z.**, Masood S., & Gwak J. (2025). Full-Resolution Transformer-Based Approach for Enhanced Teeth Segmentation and Numbering. Under-review at *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, **IF: 5.2**.
- **Ullah Z.**, Masood S., & Gwak J. (2025). Tooth Instance Segmentation in Panoramic Dental Images Using Pixel-Pair Affinity Pyramid and Cascaded Graph Partitioning. Under-review at *International Journal of Medical Informatics (IJMI)*, **IF: 4.9**.
- **Ullah Z.**, Masood S., & Gwak J. (2025). FE-Net: Correlation-based 2D feature enhancement for brain tumor segmentation. Under-review at *Engineering Applications of Artificial Intelligence (EAAI)*, **IF: 8.0**.
- Gwak J., **Ullah Z.**, Ahmad S., & Masood S. (2025) Pioneering the Path Forward: A Rigorous Analysis of Recent Developments in Transformer Models and Their Implications for the Future. Under-review at *Journal of Big Data*, **IF: 8.1**
- Masood S., **Ullah Z.**, & Gwak J. (2025). Stabilized cross-modal distillation for monocular depth estimation. Under-review at *IEEE Transactions on Pattern Analysis and Machine Intelligence*, **IF: 23.6**.

Conference Publications

Published studies

- **Ullah, Z.**, & Lee, S, H. (2019). **Magnetic Resonance Brain Image Contrast Enhancement Using Histogram Equalization Techniques**. In Journal of the Korea Society of Computer and Information.
- Ho, TKK., Jeon, Y., Na, E., **Ullah, Z.**, Kim, BC, Lee, KH., Song, JI., & Gwak, J. (2021, December). **DeepADNet: A CNN-LSTM model for the multi-class classification of Alzheimer's disease using multichannel EEG**. In 2021 DEMENTIA CARE AND PSYCHOSOCIAL FACTORS.
- Gwak, J., Kang, J., Lim, H., Min, D., & **Ullah, Z.** (2021, May). **Anomaly detection system using ResUNet++-based image restoration technique**. In 2021 conference of the Korean Society for Next-Generation Computing (pp. 401-403).
- Gwak, J., Kang, J., Lim, H., Kim, M., & **Ullah, Z.** (2021, May). **Efficient data preprocessing method for anomaly detection based on restoration model**. In 2021 conference of the Korean Society for Next-Generation Computing (pp. 375-377).

Editorial Board Member

- Guest Editor;** AI
- Guest Editor;** Diagnostic
- Editor;** Journal of Operations Intelligence
- Editor;** Spectrum of Operational Research
- Editor;** Spectrum of Decision Making and Applications
- Editor;** Journal of Decision Analytics and Intelligent Computing

Academic Services

- Reviewer of** IEEE Transactions on Medical Imaging
- Reviewer of** Information Sciences
- Reviewer of** IEEE Transactions on Artificial Intelligence
- Reviewer of** Expert Systems with Applications
- Reviewer of** Medical Physics
- Reviewer of** Neurocomputing
- Reviewer of** Biomedical Signal Processing & Control
- Reviewer of** Multimedia Tools and Applications
- Reviewer of** Scientific Reports
- Reviewer of** Sensors MDPI

Skills & Tools

Python (Tensorflow (Keras), PyTorch, VTK), MATLAB, C, C++, Numpy, Pandas, Matplotlib, Html, PHP